

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Solves to find $\tan 2\theta$ (allow M1 even if only $+\sqrt{3}$ given PI by any correct final value of θ)	AO1.1a	M1	$\tan 2\theta = \pm\sqrt{3}$ $\tan 2\theta = \sqrt{3} \rightarrow 2\theta = 60^\circ, 240^\circ, 420^\circ, 600^\circ$ $\tan 2\theta = -\sqrt{3} \rightarrow 2\theta = 120^\circ, 300^\circ, 480^\circ, 660^\circ$ $\theta = 30^\circ, 60^\circ, 120^\circ, 150^\circ, 210^\circ, 240^\circ, 300^\circ, 330^\circ$
Obtains at least 3 correct final values for θ ignore extra incorrect terms or terms outside range	AO1.1b	A1	
Obtains at least 3 correct final values of θ from $\tan 2\theta = \sqrt{3}$ and at least 3 correct values from $\tan 2\theta = -\sqrt{3}$ ignore extra incorrect terms or terms outside range	AO1.1a	M1	
Obtains complete set of exactly 8 correct values for θ	AO1.1b	A1	
			Total 4 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Demonstrates a clear understanding that $\sin x = 0$ is a solution, and that this has not been properly taken into account.	AO2.3	R1	$\sin x = 0$ leads to a solution, but when she cancelled $\sin x$ she effectively assumed it was not equal to 0 and hence lost a number of solutions.
Explains that cancelling $\sin x$ is not allowed if it is zero / only allowed if it is non-zero	AO2.4	E1	
			Total 2 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Divides or multiplies by $\cos \theta$	AO3.1a	M1	$\frac{\sin \theta \tan \theta}{\cos \theta} + 2 \frac{\sin \theta}{\cos \theta} = 3$
Obtains correct quadratic	AO1.1b	A1	$\tan^2 \theta + 2 \tan \theta - 3 = 0$
Applies a correct method to solve 'their' quadratic PI	AO1.1a	M1	$(\tan \theta + 3)(\tan \theta - 1) = 0$ $\tan \theta = 1$ or -3
Finds two correct values of $\tan \theta$ from 'their' quadratic	AO1.1b	A1F	$\theta = 45^\circ$ or 108° ALT
Obtains two correct answers CAO	AO1.1b	A1	$\sin \theta \tan \theta \cos \theta + 2 \sin \theta \cos \theta = 3 \cos^2 \theta$ $\sin^2 \theta + 2 \sin \theta \cos \theta - 3 \cos^2 \theta = 0$ $(\sin \theta + 3 \cos \theta)(\sin \theta - \cos \theta) = 0$ $\tan \theta = 1$ or -3 $\theta = 45^\circ$ or 108°
			Total 5 marks

Q4.

(a) $x + 30^\circ = 79^\circ, x + 30^\circ = 180^\circ + 79^\circ$

$x = 49^\circ$

49 as the only solution in the interval $0^\circ \leq x \leq 90^\circ$

B1

$x = 229^\circ$

AWRT 229. Not given if any other soln. in the interval $90^\circ \leq x \leq 360^\circ$.

Ignore anything outside $0^\circ \leq x \leq 360^\circ$

B1

2

(b) Translation;

*Accept 'translat...' as equivalent.
[T or Tr is NOT sufficient]*

B1

$$\begin{bmatrix} -30^\circ \\ 0 \end{bmatrix}$$

OE Accept **full** equivalent to vector in words provided linked to 'translation / move / shift' and **correct** direction. (0 / 2 if >1 transformation).

B1

2

(c) (i) $5 + \sin^2 \theta = (5 + 3\cos \theta) \cos \theta$
 $\Rightarrow 5 + \sin^2 \theta = 5\cos \theta + 3\cos^2 \theta$
 Correct RHS.

B1

$$5 + 1 - \cos^2 \theta = 5\cos \theta + 3\cos^2 \theta$$

$\sin^2 \theta = 1 - \cos^2 \theta$ used to get a quadratic in $\cos$.

M1

$$6 = 5\cos \theta + 4\cos^2 \theta \text{ or } 4\cos^2 \theta + 5\cos \theta - 6 (= 0)$$

ACF with like terms collected.

A1

$$\Rightarrow (4\cos \theta - 3)(\cos \theta + 2) (= 0)$$

Correct quadratic and $(4c \pm 3)(c \pm 2)$ or by formula
 OE PI by $\pm$ 'correct' 2 values for $\cos \theta$.

m1

Since $\cos \theta \neq -2$, $\cos \theta = \frac{3}{4}$

CSO AG. Must show that the 'soln' $\cos \theta = -2$ has been considered and rejected

A1

5

(ii) $5 + \sin^2 2x = (5 + 3\cos 2x) \cos 2x \Rightarrow \cos 2x = \frac{3}{4}$
 Using (c)(i) to reach $\cos 2x = \frac{3}{4}$ **or** finding at least 3 solutions of $\cos \theta = \frac{3}{4}$ **and** dividing them by 2.

M1

$$2x = 0.722(7...), 2\pi - 0.722(7...),$$

$$2\pi + 0.722(7...), 4\pi - 0.722(7...),$$

Valid method to find all four 'positions' of solutions.

m1

$$x = 0.361, 2.78, 3.50, 5.92$$

CAO Must be these four 3sf values but ignore any values outside the interval $0 < x < 2\pi$.

A1

3

[12]

Q5.

$$(a) \quad \frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \frac{\sec^2 x}{\sec^2 x - 1}$$

$$\sec^2 x = 1 + \tan^2 x \text{ used}$$

M1 for correct use of $\sec^2 x = 1 + \tan^2 x$ at least once or
($\operatorname{cosec}^2 x = 1 + \cot^2 x$)

M1

$$= \frac{\sec^2 x}{\tan^2 x} \text{ or } \frac{1 + \tan^2 x}{\tan^2 x}$$

$$\left(= \frac{1}{\cos^2 x \tan^2 x} \right)$$

$$= \frac{1}{\sin^2 x} \text{ or } \cot^2 x + 1$$

Shown, with no errors

A1

$$= \operatorname{cosec}^2 x$$

AG (No errors, omissions or poor notations seen)

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3

$$(b) \quad \operatorname{cosec}^2 x = \operatorname{cosec} x + 3$$

$$\operatorname{cosec}^2 x - \operatorname{cosec} x - 3 = 0$$

must have = 0

correct solution of the quadratic,
or by completing the square

B1

$$\operatorname{cosec} x = \frac{1 \pm \sqrt{13}}{2} \text{ or } (2.3... \text{ and } -1.3...)$$

$$\left(\operatorname{cosec} x = \pm \sqrt{\frac{13}{4} + \frac{1}{2}} \right)$$

PI by values for $\sin x$

M1

$$\sin x = \frac{2}{1 \pm \sqrt{13}}$$

$$B1F \text{ for } \operatorname{cosec} x = \frac{1}{\sin^2 x} \text{ seen or implied}$$

B1F

$$= 0.434 \text{ and } -0.768 \text{ (or } -0.767)$$

PI

A1

B1 for any three values correct AWRT

B1

$$x = 26^\circ, 154^\circ, -50^\circ, -130^\circ$$

B1 for all four values correct AWRT and no extras in the interval $-180 < x < 180^\circ$

B1

6

(c) $2\theta - 60^\circ = x$

where x is a written value from candidate's (b) in degrees
PI by their answer

M1

$$\theta = 43^\circ, 5^\circ$$

CSO

A1

Ignore solutions outside interval $0^\circ < \theta < 90^\circ$

2

[11]

Q6.

(a) (i) $R = \sqrt{13}$

Accept 3.6 or better

B1

$$\tan \alpha = \frac{2}{3}$$

OE

M1

$$\alpha = 33.7^\circ$$

A1

3

(ii) minimum value = $-\sqrt{13}$

Accept - 3.6 or better; ft R

B1ft

when $x - \alpha = \cos^{-1}(-1)$

NMS 0 / 2

M1

$$x = 213.7^\circ$$

Calculus used 0 / 2

A1

3

(b) (i) LHS = $\frac{\cos x}{\sin x} - 2 \sin x \cos x$

Express $\cot x - \sin 2x$ in terms of $\sin x$ and $\cos x$; ACF

M1

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

Factor out $\frac{\cos x}{\sin x}$ and $1 - 2 \sin^2 x$

m1

$$= \cot x \cos 2x$$

All correct

A1

3

Alternatives:

$$\text{RHS} = \cot x \cos 2x$$

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

*Express $\cot x \cos 2x$ in terms of $\cos x$ and $\sin x$,
 $\cos 2x$ ACF*

(M1)

$$= \frac{\cos x}{\sin x} - 2\sin x \cos x$$

$\cos 2x = 1 - 2\sin^2 x$ and multiply out and simplify.

(m1)

$$= \cot x - \sin 2x$$

All correct.

(A1)

(3)

$$\cot x(1 - \cos 2x) - \sin 2x = 0$$

$$\frac{\cos x}{\sin x} (1 - (1 - 2\sin^2 x)) - 2\sin x \cos x = 0$$

Rearrange to expression = 0 and factor out $\cot x$;

Express $\cot x$, $\cos 2x$ and $\sin 2x$ in terms of $\sin x$ and $\cos x$, ACF

(M1)

$$\frac{\cos x}{\sin x} (2\sin^2 x) - 2\sin x \cos x = 0$$

$\cos 2x = 1 - 2\sin^2 x$ used

(m1)

$$2\sin x \cos x - 2\sin x \cos x = 0$$

Simplified, with all correct

(A1)

(3)

(ii) $\cot x - \sin 2x = 0$

$$\cot x \cos 2x = 0$$

$$\cot x = 0 \quad \text{or} \quad \cos 2x = 0$$

Both equations correct

M1

$$2x = 90^\circ \quad (270^\circ)$$

Condone missing 270°

m1

$$x = 90^\circ, 45^\circ, 135^\circ$$

All correct

Alternatives:

$$\cot x - \sin 2x = \frac{\cos x}{\sin x} - 2\sin x \cos x = 0$$

$$\cos x \left(\frac{1}{\sin x} - 2\sin x \right) = 0$$

$$\cos x = 0 \quad \text{or} \quad 1 - 2\sin^2 x = 0$$

Both equations

(M1)

$$\sin x = (\pm) \frac{1}{\sqrt{2}}$$

(m1)

$$x = 90^\circ, 45^\circ, 135^\circ$$

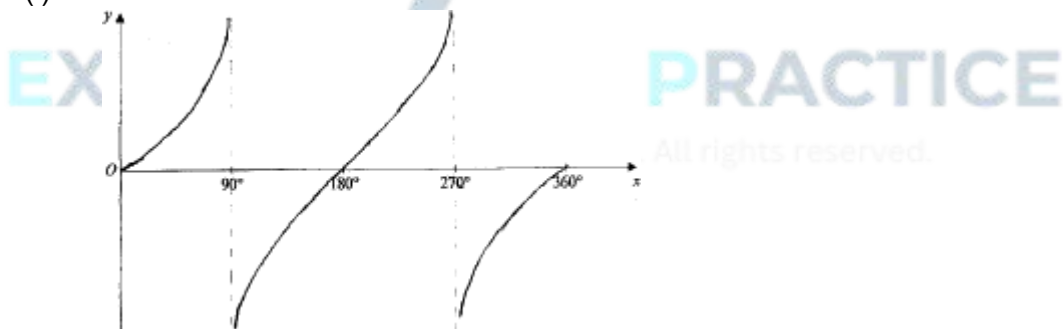
(A1)

(3)

[12]

Q7.

(a) (i)



*Ignore any part of the graph drawn outside interval
 $0^\circ \leq x \leq 360^\circ$ in (a)*

*A 3 branch curve between 0 and 360 meeting the x-axis
 at or very close to 0, 180, 360 only*

B1

*A 3 branch curve between 0 and 360 with correct shape
 tending to infinity at, at least 3, of the 4 relevant ends*

B1

Correct graph for $0^\circ \leq x \leq 360^\circ$ with correct intercepts.

Asymptotes not explicitly required but graphs should show correct 'tendency' close to 90 and 270.

B1

3

(ii) $135^\circ; 315^\circ$

B2 for both 135 and 315 and no 'extra' in interval $0^\circ \leq x \leq 360^\circ$ (If not B2 then award B1 for either 135 or 315 with or without extras)

B2,1,0

2

(c) (i) $6 \tan \theta \sin \theta = 5 \Rightarrow 6 \frac{\sin \theta}{\cos \theta} \sin \theta = 5$
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$ used

M1

$$6 \frac{\sin^2 \theta}{\cos \theta} = 5 \Rightarrow 6 \frac{1 - \cos^2 \theta}{\cos \theta} = 5$$

$\sin^2 \theta$ replaced by $1 - \cos^2 \theta$ throughout

m1

$$6 - 6\cos^2 \theta = 5\cos \theta \Rightarrow 6\cos^2 \theta + 5\cos \theta - 6 = 0$$

Completion AG Be convinced

A1

3

(ii) $6 \tan 3x \sin 3x = 5 \Rightarrow 6 \cos^2 3x + 5 \cos 3x - 6 = 0$

Using (b)(i) with $\theta = 3x$ PI by attempting to solve eg for theta then dividing soln(s) by 3

M1

$$(3\cos 3x - 2)(2\cos 3x + 3) (= 0)$$

$$(\cos 3x = 2/3, -3/2)$$

Correct factorisation or correct subst into the quadratic formula PI by two 'correct' roots

m1

$$\cos 3x = \frac{2}{3} = \cos 48.1(89..) [= \cos \alpha]$$

$$3x = \alpha, 360 - \alpha, 360 + \alpha.$$

Dep on M1 only,

$$3x = \alpha, 360 - \alpha, 360 + \alpha \text{ for c's } \alpha. \text{ from an eqn}$$

$\cos 3x = k$ where $-1 < k < 1$ OE PI and no solns
from k outside $-1 \leq k \leq 1$

m1

$$x = 16^\circ,$$

B1

$$104^\circ,$$

B1

$$136^\circ$$

B1

AWRT 16, 104, 136. Deduct one mark (from any award of these 3 B marks) if more than three solns given inside the interval $0^\circ \leq x \leq 180^\circ$.

Ignore any solutions outside the interval $0^\circ \leq x \leq 180^\circ$.

NMS Max. is B3/6

6

[14]

Q8.

$$(8\sec x - 2\sec^2 x = \tan^2 x - 2)$$

$$8\sec x - 2\sec^2 x = \sec^2 x - 1 - 2$$

Using $\tan^2 x = \sec^2 x - 1$ and NOT replacing $\sec^2 x$ with $1 + \tan^2 x$.

M1

$$3\sec^2 x - 8\sec x - 3 (= 0)$$

A1

$$(3\sec x + 1)(\sec x - 3) (= 0)$$

Correct factors or correct use of quadratic equation formula or completing the square for 'their' equation.

$$\sec x - \frac{8}{6} = \pm \sqrt{\frac{64}{36} + 1}$$

m1

$$\text{Or } \sec x = \frac{8 \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$$

$$\sec x = 3, -\frac{1}{3} \text{ (or } -0.33)$$

Both correct.

A1

$$\sec x = \frac{1}{\cos x}$$

PI

B1

$$\left(\cos x = \frac{1}{3} \text{ or } 0.33 \right)$$

$$\left(\sec x = -\frac{1}{3} \text{ is impossible} \right)$$

$$x = 1.23, 5.05$$

One correct. Must have earned A1 for correct quadratic, but independent of the second A1.

A1

Both correct and no extras in $0 < x < 2\pi$.

CAO

A1

[7]

Q9.

(a) $f\left(-\frac{3}{2}\right) = 4\left(-\frac{3}{2}\right)^3 - 11\left(-\frac{3}{2}\right) - 3$

$x = -\frac{3}{2}$ substituted

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M1

$$= -4 \times \frac{27}{8} + \frac{33}{2} - 3 = 0 \Rightarrow \text{factor}$$

Processing, = 0 and conclusion

A1

2

(b) $2x^2 - 3x - 1$

M1 for any two of a, b, c correct

M1A1

2

(c) (i) $2\cos 2\theta \sin \theta + 9\sin \theta + 3$
 $= 2(1 - 2\sin^2 \theta)\sin \theta + 9\sin \theta + 3$

$\cos 2\theta$ expanded; ACF and substituted

M1

$$= 2\sin \theta - 4\sin^3 \theta + 9\sin \theta + 3$$

All in terms of $\sin \theta$ or x and simplified to a cubic expression

m1

$$\sin \theta = x \Rightarrow 4x^3 - 11x - 3 = 0$$

*Reverse signs and express in x correctly **AG***

A1

3

$$(ii) \quad 2x^2 - 3x - 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{17}}{4}$$

Use formula correctly to solve $ax^2 + bx + c = 0$ from part (b)

M1

$$x = \frac{3 - \sqrt{17}}{4} \text{ or } -0.28\dots$$

A1

$$\theta = 196^\circ \text{ and } 344^\circ$$

Both required and no others in range; condone greater accuracy. Ignore solutions out of range.

A1

$$x = \frac{3 + \sqrt{17}}{4} \text{ no solutions for } \sin \theta$$

$$x = -\frac{3}{2} \text{ no solutions for } \sin \theta$$

Must have three correct roots and reject both other roots from cubic equation

E1

4

[11]

Q10.

$$(a) \quad 2\sin \theta = 7\cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{7}{2}$$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$ clearly used to reach either $2\tan \theta = 7$ or $2/7 \tan \theta = 1$ or $\tan \theta = 3.5$ or even $\tan \theta = 2/7$ after

$$\text{seeing } \frac{\sin \theta}{\cos \theta} = \frac{2}{7}$$

M1

$$\Rightarrow \tan \theta = \frac{7}{2}$$

$$\frac{7}{2} \text{ OE eg 3.5}$$

A1

(b) (i) $6\sin^2 x = 4 + \cos x$
 $6(1 - \cos^2 x) = 4 + \cos x$
 $\cos^2 x + \sin^2 x = 1$ used

M1

$$6 - 6\cos^2 x = 4 + \cos x$$

$$\Rightarrow 6\cos^2 x + \cos x - 2 = 0$$

CSO AG Be convinced.

A1

2

(ii) $6\sin^2 x = 4 + \cos x \Rightarrow$
 $6\cos^2 x + \cos x - 2 = 0$
 Uses (b)(i)

M1

$$(3\cos x + 2)(2\cos x - 1) (= 0)$$

$$(3c \pm 2)(2c \pm 1) \text{ or by formula}$$

M1

Correct factorisation or quadratic formula with $b^2 - 4ac$ evaluated correctly. (PI by both correct values for $\cos x$)

A1

$$\cos x = -\frac{2}{3}, \cos x = \frac{1}{2}$$

CSO Both values for $\cos x$ correct. Accept 3sf rounded or truncated.

$$x = 132^\circ, 228^\circ, 60^\circ, 300^\circ$$

B1 for any 3 of the 4 values correct. Condone greater accuracy (131.810...; 228.189...).

Ignore answers outside the given interval. Deduct 1 mark from these two B marks for each extra solution

if more than 4 answers in the given interval to a min of B0
NMS: max possible is B2

B2,1,0

6

[10]

Q11.

(a) $\sec^2 \theta - 1 = \dots$

correct use of $\sec^2 \theta = 1 + \tan^2 \theta$

B1

$\sec^2 \theta + 3\sec \theta - 10 (= 0)$

quadratic expression in $\sec \theta$ with all terms on one side

M1

$(\sec \theta + 5)(\sec \theta - 2) = 0$

*attempt at factors of their quadratic, $(\sec \theta \pm 5)(\sec \theta \pm 2)$,
or correct use of quadratic formula*

m1

$\sec \theta = -5, 2$

A1

$\left(\cos \theta = -\frac{1}{5}, \frac{1}{2} \right)$

$60^\circ, 300^\circ, 101.5^\circ, 258.5^\circ$ (AWRT)

3 correct, ignore answers outside interval

B1

all correct, no extras in interval

B1

6

(b) $4x - 10^\circ = 60^\circ, 101.5^\circ, 258.5^\circ, 300^\circ$

$4x - 10 = \text{any of their } (60),$

M1

$4x = 70^\circ, 111.5^\circ, 268.5^\circ, 310^\circ$

all their answers from (a), BUT must have scored B1

A1F

$x = 17.5^\circ, 27.9^\circ, 67.1^\circ, 77.5^\circ$ (AWRT)
CAO, ignore answers outside interval

A1

[9]

Q12.

(a) $16\left(\frac{3}{4}\right)^3 + 11\left(\frac{3}{4}\right) - 15$

Evaluate $f\left(\frac{3}{4}\right)$ not long division.

M1

$$= \frac{27}{4} + \frac{33}{4} - 15 = 0 \Rightarrow \text{factor}$$

Processing and conclusion.

A1

For A1; minimum processing seen; $16 \times \frac{27}{64} + 11 \times \frac{3}{4} - 15 = 0$; $15 - 15 = 0$

and no other working is A0 minimum conclusion = 0 hence factor

2

(b) $27\cos\theta(2\cos^2\theta - 1) +$

Use acf of $\cos 2\theta$ formula

B1

$$19\sin\theta(2\sin\theta\cos\theta) - 15 = 0$$

Use acf of $\sin 2\theta$ formula

B1

$$54\cos^3\theta - 27\cos\theta + 38(1 - \cos^2\theta)\cos\theta - 15 = 0$$

All in cosines.

M1

$$16\cos^3\theta + 11\cos\theta - 15 = 0$$

$$x = \cos\theta \Rightarrow 16x^3 + 11x - 15 = 0$$

Simplification and substitute $x = \cos\theta$ to obtain AG CSO.

A1

For M1 mark; $\cos 2\theta$ (eventually) in form $a \cos^2\theta + b$; $19\sin\theta \sin 2\theta$ in form $c \cos\theta \sin^2\theta$ and use $\sin^2\theta = 1 - \cos^2\theta$ to obtain $c \cos\theta (1 - \cos^2\theta)$

(c) $16x^3 + 11x - 15 = (4x - 3)(4x^2 + 3x + 5)$

Factorise $f(x)$

M1A1

$$b^2 - 4ac = 3^2 - 4 \times 4 \times 5 \quad (= -71)$$

Find discriminant of quadratic factor; or seen in formula

m1

$$b^2 - 4ac \quad x^2 + 3x + 5 = 0)$$

Conclusion; CSO

$$\Rightarrow \text{(only) solution is } \cos \theta = \frac{3}{4}$$

Condone $\frac{3}{4}$ is (only) solution

A1

M1 $(4x - 3)(4x^2 + kx \pm 5)$ A1 fully correct

m1 candidate's values of a, b, c used in expression for $b^2 - 4ac$

or complete square to obtain $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$

A1 $b^2 - 4ac$ correct or $\left(x + \frac{3}{8}\right)^2 = \frac{9}{64} - \frac{5}{4} \quad \left(= -\frac{71}{64}\right)$ and stated

to be negative so no solution or solutions are not real (imaginary)
Accept imaginary solutions from calculator if stated to be imaginary.

Condone $\sqrt{-71}$ is negative, or similar, so no solution.

Conclusion $x = \frac{3}{4}$ is solution, or $\cos \theta = \frac{3}{4}$ is solution.

Q13.

$$\tan \theta = -1$$

$$\sin^2 \theta = 3\cos^2 \theta$$

B1

$$\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

$\frac{\sin \theta}{\cos \theta} = \tan \theta$ used on $\sin^2 \theta - 3\cos^2 \theta$ or forms and solves
 a correct quadratic in $\sin$ or $\cos$ and then uses to find $\tan \theta$

M1

$$\tan^2 \theta = 3$$

$$\begin{aligned}
 \tan^2 \theta &= 3 \text{ or } \tan^2 \theta - 3 = 0 \\
 \text{or } (\tan \theta + \sqrt{3})(\tan \theta - \sqrt{3}) &= 0 \\
 \text{or } \tan \theta &= \sqrt{3} \text{ or } \tan \theta = -\sqrt{3}
 \end{aligned}$$

A1

$$\tan \theta = \pm \sqrt{3}$$

Both

A1

[4]

Q14.

$$\begin{aligned}
 \text{(a) LHS} &= \frac{(1 - \cos \theta) + (1 + \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)} \\
 &\quad \text{Combining fractions}
 \end{aligned}$$

M1

$$\begin{aligned}
 &= \frac{2}{1 - \cos^2 \theta} \\
 &\quad \text{Correctly simplified}
 \end{aligned}$$

A1

$$= \frac{2}{\sin^2 \theta}$$

Use of $\sin^2 \theta + \cos^2 \theta = 1$

m1

$$\begin{aligned}
 2 \operatorname{cosec}^2 \theta &= 32 \\
 \operatorname{cosec}^2 \theta &= 16
 \end{aligned}$$

AG; no errors seen

A1

OR

$$\begin{aligned}
 1 - \cos \theta + 1 + \cos \theta &= 32(1 + \cos \theta)(1 - \cos \theta) \text{ (M1)} \\
 2 &= 32(1 - \cos^2 \theta) \text{ (A1)} \\
 2 &= 32\sin^2 \theta \text{ (m1)} \\
 \operatorname{cosec}^2 \theta &= 16 \text{ (A1)}
 \end{aligned}$$

(b) $\operatorname{cosec} y = (\pm) \sqrt{16}$ or better (PI by further working)

or $\sin y = (\pm) \sqrt{\frac{1}{16}}$ or better

M1

(y =)

0.253, (2.889,) (3.394,) (6.031,) (-0.253)

Sight of any of these correct to 3dp or better

B1

(y =)

0.25, 2.89, 3.39 (or better)

Must see these 3 answers, with or without either / both of -0.25 or 6.03

Ignore answers outside interval -0.25 to 6.03 but extras in this interval scores A0

A1

$x = 0.43, 1.74, 2(.00), 0.17$

3 correct (must be 2 dp)

All 4 correct (must be 2 dp) and no extras in interval (ignore answers outside interval)

B1

B1

5

[9]

Q15.

(a) $R = \sqrt{10}$

Accept 3.2 or better. Can be earned in (b)

B1

$\tan \alpha = 3$

OE; M0 if $\tan \alpha = -3$ seen

M1

$\alpha = 71.6$ or better

$\alpha = 71.56505\dots$

A1

3

(b) $\sin(x \pm \alpha) = \frac{-2}{R}$
or their R and / or their α ; PI

M1

$x(= -39.2 + 71.6) = 32(.333)$
32 or better
Condone 32.4

A1

or

$x - 71.6 = 219.2$
must see 219 and 72 or better PI by 291 or better as answer
Condone extra solutions

m1

$x = 291$

Condone 290.8 or better CSO Withhold final A1 if more than two answers given within interval

A1

4

[7]

Q16.

(a) $\tan x = -3$

$\Rightarrow = \tan^{-1}(-3) (= -71.56\dots)^\circ$
PI eg by 71(.56..) or -71(.56..) seen

M1

$x = 108^\circ, 288^\circ$

Condone more accurate answers.

(108.4349..., 288.4349...).

[Ignore answers outside interval; If more than 2 answers inside interval -1 from A marks for each extra to a min of 0]

A1,A1

3

(b) (i) $7\sin^2 \theta + \sin \theta \cos \theta = 6(\cos^2 \theta + \sin^2 \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used; OE

M1

$$7\sin^2 \theta - 6\sin^2 \theta + \sin \theta \cos \theta - 6\cos^2 \theta = 0$$

$$\Rightarrow \sin^2 \theta + \sin \theta \cos \theta - 6\cos^2 \theta = 0$$

$$\Rightarrow \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\sin \theta}{\cos \theta} - 6 = 0$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta \text{ used}$$

M1

$$\Rightarrow \tan^2 \theta + \tan \theta - 6 = 0$$

CSO AG

A1

3

(ii) $(\tan \theta + 3)(\tan \theta - 2) = 0$

Factorise or other valid method to solve quadratic

M1

$$\tan \theta = -3 \text{ or } \tan \theta = 2$$

Need both

A1

$$\theta = 108^\circ, 288^\circ; \quad \theta = 63^\circ, 243^\circ;$$

Only ft on (a) for the c's two +ve $\tan^{-1}(-3)$ vals.

[B1 if 3 correct (ft)]

Condone more accurate answers.

(108.4349..., 288.4349...;

63.4349..., 243.4349...)

[Ignore answers outside interval; If more than 2 answers for each inside interval, -1 for each extra from Bs to a min of 0]

B2F,1F

4

[10]

Q17.

(a) $\cos x = -0.2$

$$\text{Or } \tan x = (\pm) \sqrt{24}$$

M1

$$x = 1.77, 4.51$$

AWRT

One correct value

A1

*Second correct value and no extra values
in interval 0 to 6.28...*

Ignore answers outside interval

SC

x = 1.8, 4.5 with or without working M1 A1 A0

SC (using degrees)

101.54, 281.54 M1 A1 A0

101.5, 281.5 M1 A0 A0

SC

No working shown

2 correct answers 3/3

1 correct answer 2/3

A1

3

(b) LHS

$$= \frac{\operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x)}{(1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x)}$$

*Correctly combining fractions but
condone poor use, or omission, of brackets*

M1

$$= \frac{\operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x}{1 - \operatorname{cosec}^2 x}$$

Allow recovery from incorrect brackets

A1

$$= \frac{-2 \operatorname{cosec}^2 x}{-\cot^2 x} \text{ or } \frac{-2(1 + \cot^2 x)}{-\cot^2 x}$$

*Correct use of relevant trig identity
eg $\operatorname{cosec}^2 x = 1 + \cot^2 x$*

m1

$$2 \sec^2 x = 50$$

$$\sec^2 x = 25$$

AG

*All correct with no errors seen INCLUDING
correct brackets on 1st line*

A1

4

Or

$$\frac{\operatorname{cosec} x}{1 + \operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1 - \operatorname{cosec} x} = 50$$

$$\begin{aligned} \operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x) \\ = 50 (1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x) \end{aligned}$$

*Correctly eliminating fractions but
condone poor use, or omission, of brackets*

(M1)

$$\begin{aligned} \operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x \\ = 50 (1 - \operatorname{cosec}^2 x) \end{aligned}$$

Allow recovery from incorrect brackets

(A1)

$$48 \operatorname{cosec}^2 x = 50$$

$$\sin^2 x = \frac{24}{25} \Rightarrow \cos^2 x = \frac{1}{25}$$

*Correct use of relevant trig identity
eg $\sin^2 x = 1 - \cos^2 x$*

(m1)

$$\sec^2 x = 25$$

AG

*All correct with no errors seen INCLUDING
correct brackets on 1st line*

(A1)

(c) $\sec x = \pm 5$

Or $\cos x = \pm 0.2$

Or $\tan x = \pm \sqrt{24}$

M1

$x = 1.77, 4.51, 1.37, 4.91$ (AWRT)

3 correct

A1

*4 correct and no other answers in interval
Ignore answers outside interval*

SC

1.8, 4.5, 1.4, 4.9

With or without working

M1 A1

SC

their 2 answers from (a)

+1.37, 4.91

(AWRT)

2/3

SC For this part, if in degrees

max mark is

M1 A0

SC

No working shown

4 correct answers 3/3

3 correct answers 2/3

0, 1, 2 correct answers 0/3

A1

3

[10]

Q18.

(a) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Condone numerator as $\tan x + \tan x$

B1

$$2 \tan x + \tan x (1 - \tan^2 x) = 0$$

Multiplying throughout by their denominator

M1

$$\tan x = 0$$

$$\text{or } (2 + 1 - \tan^2 x) = 0 \Rightarrow \tan^2 x = 3$$

AG Must show $\tan x = 0$ and $\tan^2 x = 3$

A1

3

Alternative

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} + \frac{\sin x}{\cos x} = 0$$

(B1)

$$2\sin x \cos^2 x + \sin x (\cos^2 x - \sin^2 x) = 0$$

$$\sin x (2\cos^2 x + \cos^2 x - \sin^2 x) = 0$$

(M1)

$$\left. \begin{array}{l} \Rightarrow \sin x = 0 \\ \Rightarrow \tan x = 0 \end{array} \right\} \text{ and } \left. \begin{array}{l} 3\cos^2 x = \sin^2 x \\ \tan^2 x = 3 \end{array} \right\}$$

(A1)

(3)

(ii) $x = 60$ **AND** $x = 120$

*Condone extra answers outside interval
eg 0 and 180*

B1

1

(b) (i) $2\sin x \cos x = \cos x f(x)$

*Where $f(x) = \cos^2 x - \sin^2 x$
or $2\cos^2 x - 1$ or $1 - 2\sin^2 x$*

M1

$$2\sin x \cos x = \cos x (1 - 2\sin^2 x)$$

A1

$(\cos x \neq 0) \quad 2\sin x = 1 - 2\sin^2 x$

$$2\sin^2 x + 2\sin x - 1 = 0$$

AG

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A1

3

(ii) $\sin x = \frac{-2 \pm \sqrt{4 - 4 \times 2 \times (-1)}}{2 \times 2}$

*Correct use of quadratic formula or completing
the square or correct factors*

M1

$$\sin x = \frac{-2 \pm 2\sqrt{3}}{4}$$

$\sqrt{12}$ must be simplified and must have $\pm$.

A1

$$\left. \begin{array}{l} \sin x = \frac{-1 - \sqrt{3}}{2} \text{ has no solution} \\ \sin x = \frac{\sqrt{3} - 1}{2} \end{array} \right\}$$

Reject one solution and state correct solution.

E1

3

[10]

Q19.

(a) $h = 0.5$

*$h = 0.5$ stated or used. (PI by x -values
0, 0.5, 1, 1.5, 2 provided no contradiction)*

B1

$$f(x) = \sin x$$

$$I \approx h/2\{...\}$$

$$\{.\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of 'trapezia'..

M1

$$\{.\} =$$

$$0 + 0.90929.. + 2[0.4794.. + 0.84147.. + 0.99749..]$$

$$\{.\} = 0.90929.. + 2[2.318..] = \{ 0.90929.. + 4.636.. \}$$

*Min. of 2dp values rounded or truncated.
Can be implied by later correct work
provided > 1 term or a single term which
rounds to 1.39*

A1

$$(I \approx) 0.25[5.546..] = 1.3865.. = 1.39 \text{ (to 3sf)}$$

CAO Must be 1.39

A1

4

(b) Stretch(I) in y -direction(II) scale factor 2(III)

Need (I) and either (II) or (III)

M1

*All correct. Need (I) and (II) and (III)
[> 1 transformation scores 0/2]*

A1

2

(c) $\frac{\sin x}{\cos x} = \frac{1}{2}$; $\tan x = 0.5$

$$\frac{\sin x}{\cos x} = \tan x \quad \text{used to get } \tan x = k$$

or identity $\cos^2 x + \sin^2 x = 1$ used to get either $\sin^2 x = p$ or to get $\cos^2 x = q$, (p and q must be between 0 and 1)

M1

$$\tan x = 0.5$$

Either $\tan x = \frac{1}{2}$ or $\cos x = \pm \sqrt{\frac{4}{5}}$ ($= \pm 0.894..$)

or $\sin x = \pm \sqrt{\frac{1}{5}}$ ($= \pm 0.447..$)

A1

$$x = \alpha \text{ or } \pi + \alpha \text{ where } \alpha = \tan^{-1}(k)$$

Correct method to find 2nd angle. Any in wrong ft quadrants then m0. In case of squaring method candidates must also have rejected the extra 'quadrants' for the m1. Condone degrees or mixture

m1

$$x = 0.464, 3.61$$

Both. Condone > 3sf [0.463(6..), 3.60(5..or 6.)]

Accept pair of truncated values [0.463, 3.60]

Ignore any answers outside interval 0 to 6.28

A1

4

[10]

Q20.

(a) (i) $(\sin^{-1} \pm 0.25) \pm 14.5$

PI by sight of 194.5 etc

condone ± 14.4

M1

$$\theta = 194.5, 345.5 \text{ (AWRT)}$$

no extras in interval, ignore answers outside interval

A1

2

(ii) $2 \cot^2 (2x + 30) = 2 - 7 \operatorname{cosec} (2x + 30)$

condone replacing $2x + 30$ by Y

$$2(\operatorname{cosec}^2 (2x + 30) - 1) = 2 - 7 \operatorname{cosec} (2x + 30)$$

correct use of $\operatorname{cosec}^2 Y = 1 + \cot^2 Y$

M1

$$2 \operatorname{cosec}^2 (2x + 30) + 7 \operatorname{cosec} (2x + 30) - 4 (= 0)$$

must be in this form

A1

$$(2 \operatorname{cosec} (2x + 30) \pm 1)(\operatorname{cosec} (2x + 30) \pm 4) (= 0)$$

attempt at factorisation

m1

$$\operatorname{cosec}(2x + 30) = \frac{1}{2} \text{ or } -4$$

must be this line using $f(2x + 30)$

A1

$$2x + 30 = 194.5, 345.5$$

$$x = 82.2, 157.8 \text{ (AWRT)}$$

one correct answer, allow 82.3, ignore extra solutions

B1

*CAO both answers correct and no extras
in interval, ignore answers outside interval*

B1

6

(b) stretch (I)

scale factor $\frac{1}{2}$ (II)

parallel to x -axis (III)

I and either II or III

M1

$$I + II + III$$

A1

translate

E1

$$\begin{pmatrix} -15 \\ 0 \end{pmatrix}$$

condone '15 to left' or '-15 in x (direction)'

B1

4

alternative method

translate

(E1)

$$\begin{pmatrix} -30 \\ 0 \end{pmatrix}$$

stretch

as above

(B1)

(M1)

scale factor $\frac{1}{2}$

parallel to x -axis

as above

(A1)

[12]

Q21.

- (a) $x + 52^\circ = (22^\circ), 180^\circ + 22^\circ; 360^\circ + 22^\circ$
 $(x = 180 + 22 - 52; x = 360 + 22 - 52)$

$x + 52 = 180 + \text{AWRT } 22, 360 + \text{AWRT } 22$ OE
 (max of M1 if extras in range)
 LHS could be any letter but not x unless
 final answer shows recovery
 Ms can be PI

M1;M1

$$x = 150^\circ, 330^\circ$$

Both CAO with no extras in $0^\circ \leq x \leq 360^\circ$

Ignore anything outside $0^\circ \leq x \leq 360^\circ$

A1

3

(b) (i) $3 \tan \theta = \frac{8}{\sin \theta} \Rightarrow 3 \frac{\sin \theta}{\cos \theta} = \frac{8}{\sin \theta}$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ used/seen}$$

M1

$$\frac{3(1 - \cos^2 \theta)}{\cos \theta} = 8$$

$$\sin^2 \theta = 1 - \cos^2 \theta \text{ used}$$

M1

$$\Rightarrow 3 - 3 \cos^2 \theta = 8 \cos \theta$$

$$\Rightarrow 3 \cos^2 \theta + 8 \cos \theta - 3$$

CSO AG Completion

A1

3

(ii) $(3 \cos \theta - 1)(\cos \theta + 3) = 0$

Any valid method to solve the quadratic

M1

$$\cos \theta = \frac{1}{3}$$

CSO Must only be the one value

A1

2

(iii) $\cos 2x = \frac{1}{3}$

Using (ii) OE to get or use $\cos 2x = k$

where $-1 \leq k \leq 1$

M1

$$(2x =) 70.528..$$

Award for $\cos^{-1} (1/3) =$ value from 70 to 71 inclusive, even if θ used. PI

B1

$$2x = 360^\circ - 70.528.. (= 289.47...)$$

$$2x = 360 - \cos^{-1}(c's k) \text{ OE}$$

No extras inside the range

m1

$$x = 35^\circ, 145^\circ \text{ (to the nearest degree)}$$

Both, condoning greater accuracy, with

no extras in $0^\circ \leq x \leq 180^\circ$

Ignore anything outside $0^\circ \leq x \leq 180^\circ$

SC for (b)(iii) only when c's answer for

$$(b)(ii) \text{ is } \cos \theta = -\frac{1}{3} :$$

max mark M1B1 (val 70-71 or val 109-110 inclusive) m1A0

A1

4

[12]

Q22.

(a) $\sin x = \frac{1}{3}$, or sight of ± 0.34 , $\pm 0.11\pi$ or ± 19.47 (or better)

M1

$$x = 0.34, 2.8(0) \quad \text{AWRT}$$

Penalise if incorrect answers in range;
ignore answers outside range

A1

2

(b) $\operatorname{cosec}^2 x - 1 = 11 - \operatorname{cosec} x$

Correct use of $\cot^2 x = \operatorname{cosec}^2 x - 1$

M1

$$\operatorname{cosec}^2 x + \operatorname{cosec} x - 12 (= 0)$$

A1

$$(\operatorname{cosec} x + 4)(\operatorname{cosec} x - 3) (= 0)$$

Attempt at Factors

Gives $\operatorname{cosec} x$ or -12 when expanded

Formula one error condoned

m1

$$\left. \begin{array}{l} \operatorname{cosec} x = -4, 3 \\ \sin x = -\frac{1}{4}, \frac{1}{3} \end{array} \right\}$$

Either Line

A1

$$\sin x = -\frac{1}{4}$$

$$\Rightarrow x = 3.39, 6.03 \quad \text{AWRT}$$

*3 correct or their two answers from (a)
and 3.39, 6.03*

B1F

$$0.34, 2.8(0) \quad \text{AWRT}$$

*4 correct and no extras in range
ignore answers outside range*

SC 19.47, 160.53, 194.48, 345.52 B1

B1

6

Alternative

$$\frac{\cos^2 x}{\sin^2 x} = 11 - \frac{1}{\sin x}$$

$$\cos^2 x = 11 \sin^2 x - \sin x$$

*Correct use of trig ratios and multiplying
by $2 \sin^2 x$*

(M1)

$$1 - \sin^2 x = 11 \sin^2 x - \sin x$$

$$0 = 12 \sin^2 x - \sin x - 1$$

(A1)

$$0 = (4 \sin x + 1)(3 \sin x - 1)$$

Attempt at factors as above

(m1)

$$\sin x = -\frac{1}{4}, \frac{1}{3}$$

(A1)

As above

(B1F)(B1)

[8]

Q23.

(a) $R = \sqrt{10}$

Accept $R = 3.16$ or better

B1

$\tan \alpha = 3$

OE

M1

$\alpha = 1.249$ ignore extra out of range

AWRT 1.25 SC $\alpha = 0.322$ B1
radians only

A1

3

(b) (i) minimum value = $-\sqrt{10}$
 F on R

B1F

1

(ii) $\cos(x - \alpha) = -1$

M1

$x = 4.391$

AWRT 4.39
 51.56° or ... 57° or better

A1F

2

(c) $\cos(x - \alpha) = \frac{2}{\sqrt{10}}$

M1

$x - \alpha = \pm 0.886$ 5.397

ignore extra out of range

*Two values, accept 2dp and condone 5.4
condone use of degrees*

A1

$$x = 0.36296.. \quad 2.13512..$$

F on $x - \alpha$, either value. AWRT

A1F

$$x = 0.363 \quad 2.135$$

CSO

3dp or better

A1

Alternative

$$10 \sin^2 x - 12 \sin x + 3 = 0$$

*Or equivalent quadratic using $\cos x$
(ie $\sin^2 x + \cos^2 x = 1$ used)*

M1

$\sin x =$ two numerical answers

Or equivalent using $\cos x$

A1F

$$-1 \leq \text{ans} \leq 1$$

$x =$ one correct answer

A1F

$$x = 0.363 \quad 2.135$$

CSO

3 dp or better

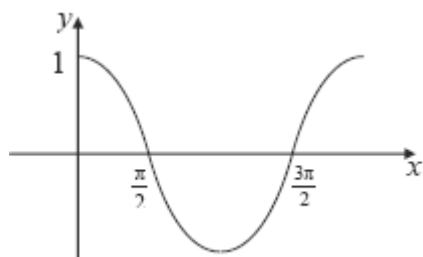
A1

4

[10]

Q24.

(a)



*Correct shape meeting positive y-axis
and only one oscillation within interval
0 to 2π*

B1

*The three correct intercepts stated;
Accept 1.57 for $\pi/2$ and 4.71 for $3\pi/2$
but must be evidence of radian vals.
not just degrees*

B1

*Ignore any parts of the graph clearly
indicated as outside the given interval*

2

(b) (i) $1 - \cos^2 \theta = \cos \theta(2 - \cos \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used

M1

$$1 = 2 \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$$

CSO AG Completion

A1

2

(ii) $\sin^2 2x = \cos 2x(2 - \cos 2x) \Rightarrow \cos 2x = \frac{1}{2}$
 Uses (b)(i)

M1

$$\{2x =\} \cos^{-1}\left(\frac{1}{2}\right) = 1.04(7..)$$

PI Accept 1.05, $\frac{\pi}{3}$; Condone 60°

m1

$$x = 0.524, 2.62$$

$$x = 0.523(59..), 2.61(7..)$$

*Condone > 3sf; Condone $x = 0.525, 2.62$
Accept truncated '3sf' vals $x = 0.523, 2.61$
Deduct 1 mark for each extra (> 2 solns) in
the given interval from A marks to a min
of A0. Ignore any solns outside the given
interval 0 to π .*

Accept, as equivalent, the exact answers

$x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$ when seen and apply
 ISW if 'errors' converting these to decimals.

If not A2 then A1 if

- one soln correct.
- 30° , 150° ie solns left in degrees
- AWRT 0.52, 2.6 ie correct vals to only 2sf.

Must see an indication that (b)(i) has been
 used otherwise 0/4 so just stating the two
 correct answers with nothing else scores 0/4.

A2,1,0

4

[8]

Q25.

(a) $(y =) 1$

Condone 1 marked at A, A = 1 etc

but not $\frac{1}{\cos 0}$, sec 0

B1

1

(b) $\cos x = \frac{1}{2}$ or $\cos^{-1} \frac{1}{2}$ seen

or sight of $\pm 60^\circ$ or $\pm \frac{\pi}{3}$, ± 1.05 (AWRT)

M1

$x = 60^\circ, 300^\circ$

Condone extra values outside

$0^\circ < x < 360^\circ$, but no extras in interval

A1

2

[3]

Q26.

(a) (i) $y = \ln(5x - 2)$

$$\frac{k}{5x - 2}$$

M1

$$\left(\frac{dy}{dx}\right) = \frac{5}{5x-2}$$

No ISW, eg $\frac{5}{5x-2} = \frac{1}{x-2}$ (M1A0)

A1

2

(ii) $y = \sin 2x$

$$\left(\frac{dy}{dx}\right) = 2 \cos 2x$$

A1

2

$$k \cos 2x$$

M1

(b) (i) $f(x) \geq \ln 0.5$ or $f(x) \geq -\ln 2$

M1A1

2

(ii) $(gf(x) =) \sin [2 \ln (5x - 2)]$
or $(gf(x) =) \sin \ln (5x - 2)^2$

Condone

$\sin 2 \ln (5x - 2)$ or $\sin 2 (\ln (5x - 2))$

but **not** $\sin 2 (\ln 5x - 2)$ or $\sin 2 \ln 5x - 2$

B1

1

(iii) $gf(x) = 0$

$$\sin[2 \ln(5x - 2)] = 0$$

$$2 \ln(5x - 2) = 0$$

Correct first step from their (b)(ii)

M1

$$5x - 2 = 1$$

Their $f(x) = 1$ from $k \ln (f(x)) = 0$

m1

$$x = \frac{3}{5}$$

Withhold if clear error seen other than omission of brackets

A1

3

- (iv) $x = \sin 2y$
 $\sin^{-1} x = 2y$ (or $\sin^{-1} y = 2x$)
Correct equation involving $\sin^{-1}$

M1

$$(g^{-1}(x)) = \frac{1}{2} \sin^{-1} x$$

A1

2

[12]

Q27.

- (a) $10 \operatorname{cosec}^2 x = 16 - 11 \cos x$
 $10(1 + \cot^2 x) = 16 - 11 \cot x$
 $10 \cot^2 x + 11 \cot x - 6 = 0$

AG Must see evidence of correct identity
 and no errors.

B1

1

- (b) Attempt at factors, giving $\pm 10 \cot^2 x \pm 6$
 when expanded.

Use of formula: condone one error

M1

$$(5 \cot x - 2)(2 \cot x + 3) (= 0)$$

Correct factors

A1

$$\left(\cot x = \frac{2}{5}, -\frac{3}{2} \right)$$

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned
Condone AWRT – 0.67
ISW if x values attempted

A1,A1

Alternative 1

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

$$10 \frac{\cos^2 x}{\sin^2 x} + 11 \frac{\cos x}{\sin x} - 6 = 0$$

$$10 \cos^2 x + 11 \cos x \sin x - 6 \sin^2 x = 0$$

$$(5 \cos x - 2 \sin x)(2 \cos x + 3 \sin x) (= 0)$$

Attempt at factors, gives

$\pm 10 \cos^2 x \pm 6 \sin^2 x$ when explained

(M1)

As above

(A1)

$$(5 \cos x = 2 \sin x \quad 2 \cos x = -3 \sin x)$$

$$\frac{5}{2} = \tan x \quad -\frac{2}{3} = \tan x$$

1st A1 must be earned

Condone AWR T -0.67

ISW if x values attempted

(A1),(A1)

Alternative 2

$$10 + 11 \tan x - 6 \tan^2 x = 0$$

$$(5 - 2 \tan x)(2 + 3 \tan x) (= 0)$$

Attempt at factors gives $\pm 10 \pm 6 \tan^2 x$

(M1)(A1)

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned

Condone AWR T -0.67

ISW if x values attempted

(A1),(A1)

4

[5]

Q28.

(a) (i) $\cos 2x = 1 - 2 \sin^2 x$

ACF in terms of $\sin$ (PI later)

B1

$$3(1 - 2 \sin^2 x) + 2 \sin x + 1 = 0$$

Substitute candidate's $\cos 2x$ in terms of $\sin x$ (at least 2 terms)

M1

$$-6 \sin^2 x + 2 \sin x + 4 = 0$$

$$3 \sin^2 x - \sin x - 2 = 0$$

AG

A1

3

(ii) $(3 \sin x + 2)(\sin x - 1) = 0$

Factorise correctly or use formula correctly

M1

$$\sin x = -\frac{2}{3} \quad \sin x = 1$$

Both; condone -0.67 or -0.66 or better

A1

2

(b) (i) $R = \sqrt{13}$

Accept 3.6 or better

B1

$$\tan \alpha = \frac{2}{3} \quad \alpha = 33.7$$

OE; accept $\alpha = 33.69(0)$

M1A1

3

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(ii) $2x - \alpha = \cos^{-1}\left(\frac{-1}{R}\right)$

Candidate's R. Or $\cos(2x - \alpha) = \frac{-1}{R}$

M1

$$2x - \alpha = 106.1^\circ, 253.9^\circ$$

$$x = 69.9^\circ, 143.8^\circ$$

One correct answer

A1

Both correct, no extras in range

A1

3

[11]

Q29.

(a) $\{x = \} \sin^{-1}(0.8) = 0.927(29\dots) \{= \beta\}$
 $\sin^{-1}(0.8) \text{ PI}$

M1

$\{x = \} \pi - \beta$

m1

$x = 0.927(29\dots), \quad 2.21 \text{ (42\dots)}$

Both

Ignore values outside interval $0 - 2\pi$ but

A0 if 'extra' values inside the given interval

A1

3

(b) (i) $\left(\frac{3\pi}{2}, -1\right)$

B1 if one coordinate correct or $\left(-1, \frac{3\pi}{2}\right)$

B2,1

2

(ii) $\pi - \alpha$

B1

1

(iii) $RS = (2\pi - \alpha) - (\pi + \alpha)$

OE eg $RS = PQ = (\pi - \alpha) - \alpha$

M1

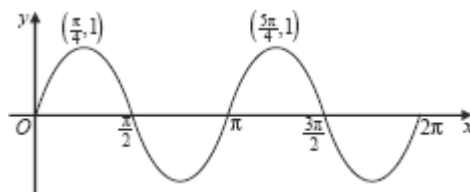
$= \pi - 2\alpha$

Must be simplified

A1

2

(c)



*Sine curve with positive gradient at O
 with at least 3 stationary points between O*

and 2π

B1

Correct shaped curve with 2 max and 2 min between 0 and 2π

B1

All 5 correct points of intersection with

x -axis with $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$ clearly shown

B1

Maximum points $\left(\frac{\pi}{4}, 1\right)$ and $\left(\frac{5\pi}{4}, 1\right)$
 stated or clearly shown on the sketch

B1 for either:

1 as the y -coordinate of max pt(s) or:
 two max pts between 0 and 2π with
 correct x -coordinates

B2,1

5

[13]

Q30.

(a) $\sec x = \frac{3}{2}$
 $\cos x = \frac{2}{3}$
 $x = 48, 312$

1 correct

B1

(Condone answers rounding to)

2 correct and no extras in interval

B1

2

(b) $2 \tan^2 x = 10 - 5 \sec x$
 $2 (\sec^2 x - 1) = 10 - 5 \sec x$

Use of trig identity correctly

M1

$2 \sec^2 x + 5 \sec x - 12 (= 0)$

A1

$$(2 \sec x - 3)(\sec x + 4) = 0$$

Attempt to solve or factorise
1 slip using formula

m1

$$\left. \begin{array}{l} \sec x = \frac{3}{2}, -4 \\ \cos x = \frac{2}{3}, -\frac{1}{4} \end{array} \right\} \text{either of these}$$

A1

$$x = 48, 312, 104, 256$$

AWRT 3 correct condone 105 or 255

B1

All correct and no extras in interval

B1

Alternative:

$$\left. \begin{array}{l} \frac{2 \sin^2 x}{\cos^2 x} = 10 - \frac{5}{\cos x} \\ 2 \sin^2 x = 10 \cos^2 x - 5 \cos x \\ 2 - 2 \cos^2 x = 10 \cos^2 x - 5 \cos x \end{array} \right\}$$

(M1)(A1)

$$12 \cos^2 x - 5 \cos x - 2 = 0$$

then rest of scheme as above

6

[8]

Q31.

(a) $\sin 2x = 2 \sin x \cos x$

OE, eg $\sin x \cos x + \sin x \cos x$ etc

B1

$$\cos x = 0 \quad x = 90, 270$$

Both required

B1

1

(b) $10 \sin x + 3 = 0$

M1

$x = 197.5 \quad 342.5$

CAO

if extra values in given range, max ½

A1A1

4

(c) $\cos 2x = \cos^2 x - \sin^2 x$

cos 2x in any correct form

B1

$2 \sin x \cos x + 1 - 2 \sin^2 x = 1 + \sin x$

sin 2x expanded and cos 2x in terms of sin x used

M1A1

$2 \sin x (\cos x - \sin x) = \sin x$

$2 (\cos x - \sin x) = 1$

CSO; need to see sin x taken out as factor or cancelled

A1

4

[9]

Q32. EXAM PAPERS PRACTICE

(a) $\frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\cos \theta} = 4$

$\tan \theta - 1 = 4$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$ *stated or used*

M1

$\tan \theta = 5$

AG; CSO

A1

2

(b) (i) $2 \cos^2 x - \sin x = 1$
 $2(1 - \sin^2 x) - \sin x = 1$

Use of $\cos^2 x + \sin^2 x = 1$

M1

$$2 - 2 \sin^2 x - \sin x = 1$$

$$\Rightarrow 2 \sin^2 x + \sin x - 1 = 0$$

AG; CSO

A1

2

- (ii) $(\sin x + 1)(2 \sin x - 1) = 0$
Factorisation or use of formula; PI by both correct values for $\sin x$

M1

$$\sin x = -1, \sin x = 0.5$$

Need both

A1

$$(\sin x = -1) \text{ so } x = 270^\circ$$

B1

$$(\sin x = 0.5) \text{ so } x = 30^\circ$$

30° as the only acute angle

A1

$$x = 180 - 30 = 150^\circ$$

*ft for 2nd angle from c's $\sin x = \text{non-integer}$
 Ignore values outside interval $0^\circ - 360^\circ$ but
 extras inside interval lose the corresp. B,
 A or B1F mark. If using rads, accepting
 either equivalent exact vals (in terms of
 π) or 2 dp values instead of degrees,
 penalise max of 1 mark from any of the
 final three marks (B1A1B1F) awarded
 NMS: 270° (B1); 30° , 150° (B1) [max 2/5]*

B1F

5

[9]

Q33.

(a) $\tan^{-1}\left(-\frac{1}{3}\right) = -0.32$

Sight of ± 0.32 or 18.43

M1

$$x = 2.82, 5.96$$

a correct answer AWR T

A1

−1 for any extra in range, ignore extra answers not in range.

[SC 161.57, 341.57 AWR T M1A1 (max 2/3)]

A1

3

(b) $3(\tan^2 x + 1) = 5 \tan x + 5$

$$3 \tan^2 x - 5 \tan x - 2 = 0$$

AG

B1

1

(c) $(3 \tan x + 1)(\tan x - 2) = 0$

Attempt at factorisation/formula

M1

$$\tan x = 2, -\frac{1}{3}$$

A1

$x = 1.11, 4.25, 2.82, 5.96$ AWR T

3 correct [SC $x = 1.11, 4.25$ + their two answers from (a)]

B1

*4 correct, no extras in range
[SC 161.57, 341.57, 63.43, 243.43
AWR T B1 (max 3/4)]*

B1

4

[8]

Q34.

(a) (i) $\cos 2x = 2 \cos^2 x - 1$

Seen in question, in consistent variable

B1

$$3(2 \cos^2 x - 1) + 7 \cos x + 5$$

Substitute candidate's $\cos 2x$ in terms of $\cos x$

M1

$$6 \cos^2 x + 7 \cos x + 2 (= 0)$$

A1

3

(ii) $(2 \cos x + 1)(3 \cos x + 2)$

*Attempt factors; formula
(‘a’ and ‘c’ correct; allow one slip)*

M1

$$\cos x = -\frac{1}{2} \quad \cos x = -\frac{2}{3}$$

Accept $-0.5, -0.67$

$$x = \cos^{-1}\left(-\frac{1}{2}\right); \cos^{-1}\left(-\frac{2}{3}\right)$$

A1

2

(b) (i) $R = \sqrt{58}$

Accept 7.6 or better

B1

$$\alpha = \sin^{-1}\left(\frac{3}{\text{their } R}\right)$$

$$\text{OE } \alpha = \sin^{-1}\left(\frac{3}{7}\right)$$

M1

$$= 23.2^\circ$$

AWRT 23.2° ($23.1985\dots$)

A1

3

(ii) $\alpha + \theta = \sin^{-1}\left(\frac{4}{\text{their } R}\right)$

Candidate's R, α

M1

$$\theta = 8.5^\circ$$

F on α , AWRT, condone 8.6

A1F

$$\theta = 125.1^\circ$$

Two solutions only, but ignore out of range

A1

3

(c) (i) $h^2 = 1 + (2\sqrt{2})^2$

Pythagoras with h or $\sec x$

M1

$$h = 3 \Rightarrow \cos \beta = \frac{1}{3}$$

AG

A1

2

(ii) $\sin 2\beta = 2 \sin \beta \cos \beta$

m1

$$\sin 2\beta = \frac{4}{9}\sqrt{2}$$

CSO; accept $p = \frac{4}{9}$ (not 0.444...)

A1

2

[15]

Q35.

(a) $\frac{3 + \sin^2 \theta}{\cos \theta - 2} = 3 \cos \theta$

$$\Rightarrow \frac{3 + (1 - \cos^2 \theta)}{\cos \theta - 2} = 3 \cos \theta$$

$\cos^2 \theta + \sin^2 \theta = 1$ stated or used [If cand starts with $\cos \theta = -\frac{1}{2}$ and gets $\sin^2 \theta = \frac{3}{4}$ without explicitly finding value for θ and verifies 1st equation is true, award M1 mo A0]

M1

$$\Rightarrow \frac{4 - \cos^2 \theta}{\cos \theta - 2} = 3 \cos \theta$$

$$\Rightarrow \frac{(2 - \cos \theta)(2 + \cos \theta)}{\cos \theta - 2} = 3 \cos \theta$$

*Difference of two squares
or division (PI by next line)*

m1

$$\Rightarrow -1(2 + \cos \theta) = 3 \cos \theta$$

A1

$$\Rightarrow -2 = 4 \cos \theta \Rightarrow \cos \theta = -\frac{1}{2}$$

CSO AG

A1

Alternative:

$$3 + 1 - \cos^2 \theta = 3 \cos^2 \theta - 6 \cos \theta$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

(M1)

$$(4 \cos \theta + 2)(\cos \theta - 2) = 0$$

Factorising or formula

(m1)

$$\cos \theta - 2 \neq 0$$

Indicates rejection of $\cos \theta = 2$

(A1)

$$\Rightarrow 4 \cos \theta = -2 \Rightarrow \cos \theta = -\frac{1}{2}$$

AG Be convinced

(A1)

4

(b) $\theta = 3x \Rightarrow \cos 3x = -\frac{1}{2}$

*Uses part (a) to reach either $\cos 3x = -0.5$
or $\cos 3x = 0.5$*

M1

$$\cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ$$

Or $\cos^{-1}(0.5) = 60^\circ$ Condone radians here

m1

$$3x = 120^\circ, 240^\circ, 480^\circ, \dots$$

$$x = 40^\circ, 80^\circ, 160^\circ$$

A1 For at least two correct.

A2, 1, 0

If > 3 solutions in the interval

$0^\circ < x < 180^\circ$, deduct 1 mark from any A marks for each extra solution.

Deduct 1 mark from any A marks if answers in radians. Ignore extra values outside the given interval.

4

[8]



EXAM PAPERS PRACTICE

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